

Spatiotemporal Assessment of Urban Thermal Discomfort in Kolkata, India: Insights From Cloud-Based Remote Sensing

Nandini Halder¹, Manoj Kumar², Akshay Deepak³, Deepak Kumar³, Naushin Yasmin, Safi Ullah, and Sami G. Al-Ghamdi⁴

I. INTRODUCTION

Abstract—Urbanization and climate change are intricately linked, significantly influencing local and regional thermal environments. Kolkata, a rapidly expanding metropolitan city in India, has witnessed substantial shifts in urban thermal dynamics due to increasing land surface temperatures (LST), the urban heat island (UHI) effect, and heightened thermal discomfort. This article integrates high-resolution remote sensing data and a cloud-based platform via Google Earth Engine to assess spatiotemporal changes in thermal discomfort in Kolkata. The article employs the urban thermal field variance index (UTFVI) to estimate the urban thermal discomfort, while the Sen's slope estimator and the modified Mann-Kendall tests are applied to assess long-term spatiotemporal trends in urban thermal conditions. Findings reveal that LST, UHI, and UTFVI are significantly increasing at the rates of 0.149 °C/year, 0.041 °C/year, and 0.0041 °C/year, respectively. The results further reveal that those areas with low vegetation cover experience extreme thermal stress, highlighting the critical role of urban greenery in mitigating heat-related discomfort. The article offers data-driven insights into Kolkata's urban thermal landscape, with guidance for policymakers in developing sustainable urban planning and climate adaptation strategies, such as expanding green spaces, implementing cool roof technologies, and enhancing urban ventilation. By leveraging cloud-based remote sensing, this article provides a scalable framework for assessing and addressing urban thermal discomfort in other rapidly urbanizing cities worldwide.

Index Terms—India, Kolkata, modified Mann-Kendall (mMK), Sen's slope estimator (SSE), urban heat island (UHI), urban thermal field variance index (UTFVI).

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THE global landscape is undergoing a rapid transformation characterized by unprecedented urbanization [1], [2], [3]. The world's 50%–55% population lives in urban areas, a proportion that is expected to increase to 68%–70% by 2050 [4], [5], [6]. Cities are major contributors to the global carbon footprint, accounting for approximately 70% of worldwide carbon dioxide emissions [7], [8]. Unplanned urban sprawl and high population density, coupled with the urban heat island (UHI) effect, exacerbate local warming and affect residents' well-being [9], [10]. The increasing frequency of extreme heat events has been linked to urbanization patterns that reduce natural vegetation and increase built-up areas, leading to significant urban thermal discomfort, particularly in suburban areas experiencing rapid land-use changes [11], [12], [13], [14].

Urban thermal discomfort is assessed using various thermal indices. Each was designed to evaluate human heat stress in different environmental conditions [15]. Common indices include the universal thermal climate index (UTCI), wet-bulb globe temperature (WBGT), wet-bulb temperature, heat index, net effective temperature, temperature-humidity index, UHI effect, and urban thermal field variance index (UTFVI) [16], [17], [18], [19], [20]. Among these indices, UTFVI provides more robust urban-scale thermal assessments [21], [22]. It provides essential insights into how urban infrastructure, land cover changes, and anthropogenic activities contribute to localized heating [23]. This index is particularly useful because it integrates land surface temperature (LST) with vegetation indices to assess the ecological impact of urban heat [24]. Unlike UTCI or WBGT, which focus on human heat stress alone, UTFVI evaluates both human thermal discomfort and environmental degradation, making it a superior choice for urban climate studies and sustainable planning [25]. Research suggests that UTFVI is more effective in identifying heat-stressed areas within cities, providing actionable data for urban planners and policymakers [26], [27].

The term ecological thermal stress broadly captures the effects of elevated surface temperatures on urban ecosystems, such as vegetation and soil degradation [28], while urban thermal discomfort is more closely associated with human health and spatial vulnerability [29]. In this article, we used urban thermal discomfort as the primary term to encompass both ecological and human impacts of surface heat, unless a contextual distinction is required.

Urban heat stress and thermal discomfort have been the focus of numerous studies across Southeast and South Asia in relation to urbanization and climate change [30], [31], [32]. Studies leveraging Google Earth Engine (GEE) and high-resolution Landsat data have successfully mapped the spatial variations of urban thermal discomfort in these countries, allowing researchers to establish correlations between green space reduction and temperature rise [33]. While extensive research has been conducted on UHI analysis and LST trends, there has been limited application of the UTFVI for detailed statistical thermal discomfort assessments in cities, like Jakarta (Indonesia) and Dhaka (Bangladesh) [34], [35], [36], [37]. Although some studies in Bangladesh, including in Dhaka, have quantified urban thermal stress using UTFVI [38], [39], [40], most have been restricted to single-year assessments or lacked integration with long-term temporal trend analysis and spatial statistical validation [41]. As such, UTFVI's full analytical potential, especially its application over multidecadal timescales and in cloud-based geospatial platforms, remains underexplored in Bangladesh and is even more limited in India [42], [43], [44]. Beyond South Asia, several studies have applied UTFVI or related methodologies in other geographic contexts. For instance, Najafzadeh et al. [21] assessed the spatiotemporal evolution of thermal stress using UTFVI in Tehran, Iran, highlighting how rapid urbanization exacerbates ecological degradation. In Turkish cities, Degerli and Cetin [22] employed UTFVI to evaluate the climatic impacts of increasing impervious surfaces and the loss of urban greenery. In China, Chen et al. [24] demonstrated the utility of LST–NDVI relationships similar to UTFVI for detecting urban thermal anomalies [45], while Weng et al. [27] conducted a pioneering investigation into surface temperature–vegetation interactions in Indianapolis, USA, underscoring the ecological implications of urban land cover patterns [24], [27]. In Saudi Arabia, UTCI was applied to assess thermal discomfort trends across Dammam, Jeddah, Madinah, Makkah, and Riyadh between 1980 and 2022, revealing increases of 0.30 °C–0.54 °C alongside rising frequency and intensity of heat stress events [46], [47]. Additionally, studies in South Korea, Brazil, and parts of Europe have incorporated UTFVI in urban climate research to monitor seasonal heat risk and vegetation-related thermal mitigation [48], [49], [50]. Despite these international advancements, the method remains underutilized for long-term trend analysis using statistically robust techniques, such as Sen's slope estimator (SSE) and the modified Mann-Kendall (mMK) test. The lack of statistical validation of UTFVI-based thermal discomfort trends using nonparametric methods and geospatial modeling has limited its utility for urban climate planning in Indian cities [51], [52], [53], [54]. Recent studies have applied UTFVI to evaluate urban heat stress across various cities in India. For example, UTFVI has been used in the Chotanagpur Plateau Region to study land use land cover (LULC) and ecological degradation [39]. In eastern India, researchers analyzed LST patterns in New Town, Kolkata, but did not apply UTFVI or multidecadal trend diagnostics [55]. Similarly, Rana et al. [53] explored the interactions between UHI and NDVI in West Bengal using machine learning, but without UTFVI integration.

This article addresses this gap by combining UTFVI with nonparametric trend analysis and GEE-based spatial modeling, offering a unique methodological contribution to urban thermal discomfort research. In Kolkata, recent remote sensing studies have made valuable contributions by analyzing LST dynamics, UHI patterns, and vegetation cover change, though UTFVI remains underutilized [56], [57], [58], [59]. Despite growing concerns about rising heat stress levels in densely populated urban centers, such as Delhi, Mumbai, and Kolkata, there is no comprehensive study that integrates UTFVI with high-resolution remote sensing data and GEE for assessing long-term urban thermal discomfort [60].

This research gap highlights the need for cloud-based remote sensing data and advanced analytical techniques to assess spatiotemporal trends in urban thermal discomfort in Kolkata, India. The present article makes a novel contribution by integrating NDVI, LST, UHI, and UTFVI over a 30-year period (1995–2024) within the city. Unlike previous articles that rely on moderate-resolution data, this approach improves the accuracy of urban climate assessment and provides deeper insights into the dynamics of spatiotemporal thermal discomfort. Notably, the study introduces UTFVI, which is a relatively unexplored metric in the context of Kolkata, to assess urban thermal stress. The study also applies an elaborate statistical framework, using SSE and the mMK tests, offering a level of temporal trend analysis that has not been extensively undertaken in earlier research. The study holds significant scientific and societal value by advancing the understanding of urban thermal environments in a rapidly urbanizing and climate-vulnerable megacity, like Kolkata. It also carries direct policy implications, offering data-driven guidance for urban planning interventions such as expanding green spaces, implementing heat-mitigation technologies, and improving building regulations to reduce heat stress and associated public health risks in Kolkata and similar urban regions.

II. STUDY AREA

Kolkata Municipal Corporation (Kolkata) administers an area of 203.93 km², situated between 22°28'00" and 22°37'30" North latitude and 88°17'30" and 88°25'00" East longitude [61]. This region has developed primarily as a linear urban area, expanding north–south along the eastern bank of the Hooghly River [62]. According to the Census of India (2011), Kolkata had a population of approximately 4.5 million, resulting in a very high population density of around 24 252 persons per km². The city experiences a tropical wet and dry climate, characterized by annual rainfall between 1400 and 1600 mm, with monthly temperatures ranging between 10 °C and 40 °C [63]. Kolkata faces significant challenges, including air pollution, poverty, substantial migration, a rising slum population comprising approximately 31% of its inhabitants, severe traffic congestion, and multiple socioeconomic issues due to rapid, unplanned urban expansion [64]. Fig. 1 shows the LULC of the study area for the years 2024 and 1995. The LULC displays different land use classifications for the selected years, including croplands, built-up areas, forests, and water bodies, highlighting agricultural plains, transport corridors, and riparian wetlands.

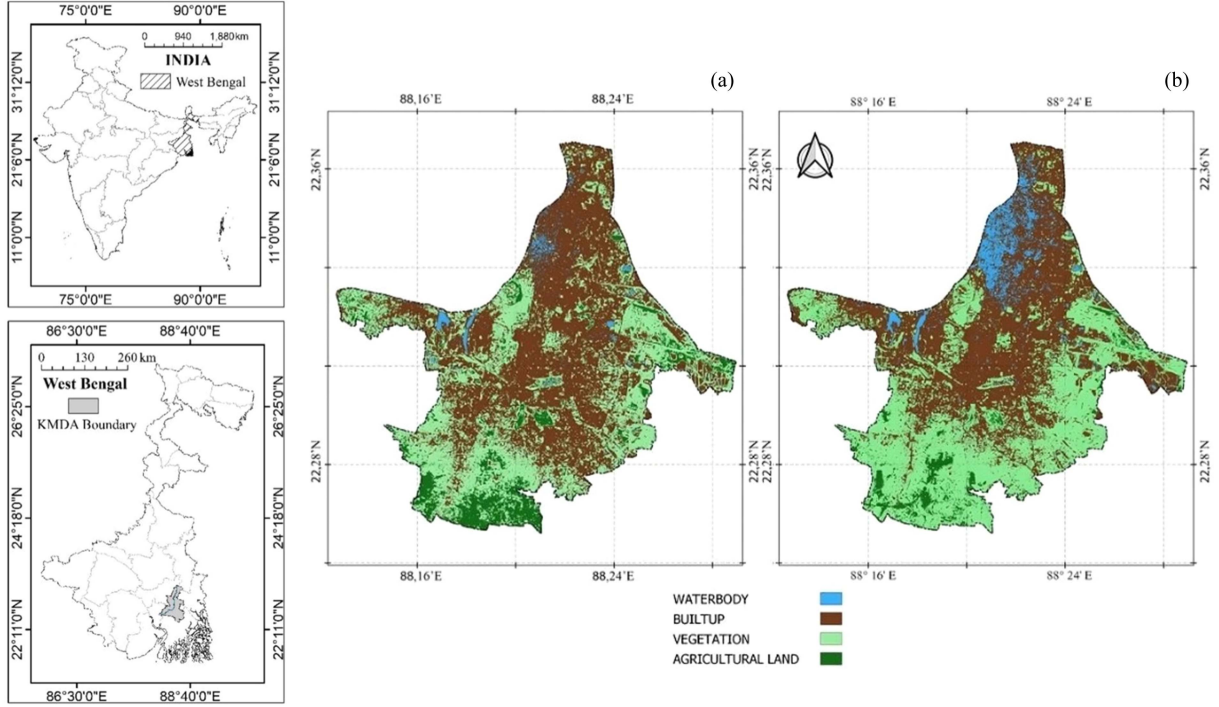


Fig. 1. Study area map with land use land cover (LULC) map of Kolkata for (a) 2024 and (b) 1995.

TABLE I
DETAILS OF THE SATELLITE DATASETS EMPLOYED IN THE STUDY FOR
PATH-ROW 141/42 FROM 1994 TO 2024, INDICATING THE SPATIAL
RESOLUTION OF VNIR (VISIBLE AND NEAR-INFRARED) AND TIR (THERMAL
INFRARED) BANDS ACROSS THREE TIME PERIODS

Year	Path-row	Band resolution - VNIR (m)	Band resolution - TIR (m)
1994–2004	141/42	30	120
2005–2014	141/42	30	100,120
2015–2024	141/42	30	100

III. DATA AND METHODS

A. Datasets

This article uses high-resolution remote sensing data from the United States Geological Survey (USGS) to assess the spatiotemporal changes in thermal discomfort in Kolkata over the period 1995–2024, utilizing the GEE platform. GEE is a cloud-based, open-source platform that facilitates large-scale remote sensing data analysis and visualization without requiring extensive local computational resources. The analysis integrates high-resolution imagery from the Landsat-5 thematic mapper (TM) (1995–2011), Landsat-7 enhanced thematic mapper plus (ETM+) (1999–2021), and Landsat-8 operational land imager/thermal infrared (TIR) sensor (2013–2024). Further details about the employed datasets are provided in Table I.

To ensure spatial consistency, all TIR bands, originally acquired at 100 or 120 m resolution, depending on the sensor, were resampled to 30-m resolution using bilinear interpolation prior to formal analysis. This resampling allowed uniform processing of LST, UHI, and UTFVI metrics across time. For

each year, monthly median composites were generated for April to September, capturing the high-temperature period typical of eastern Indian summers. The dataset includes all available cloud-free Landsat scenes for each month, and cloud-masking was performed using the Fmask algorithm embedded in the GEE environment. This approach ensures temporal consistency and minimizes noise from atmospheric effects or cloud contamination. For consistency and relevance to thermal discomfort, the study focuses exclusively on the summer months (April to September), as these months typically experience the highest thermal stress in India and the study region [65], [66].

B. Methods

To assess urban thermal discomfort and its spatiotemporal dynamics, various remote-sensing-based indices, including NDVI, LST, UHI, and UTFVI, are employed. In addition, different statistical techniques, such as SSE and mMK tests, are applied to detect significant trends in urban thermal variations in Kolkata, India. The details of these indices and statistical techniques are provided in the following sections.

1) *Calculation of NDVI, Vegetation Fraction, and Emissivity:* NDVI is utilized to quantify vegetation cover, playing a critical role in mitigating urban heat stress. Once the median composite is established, the NDVI is computed using the relevant spectral bands (NIR and red).

For Landsat 5/7, the NDVI is calculated as follows:

$$\text{NDVI} = \frac{\text{SR_B4} - \text{SR_B3}}{\text{SR_B4} + \text{SR_B3}} \quad (1)$$

where SR_B3 and SR_B4 denote shortwave bands 3 and 4, respectively.

For Landsat 8, the NDVI is calculated as follows:

$$\text{NDVI} = \frac{\text{SR_B5} - \text{SR_B4}}{\text{SR_B5} + \text{SR_B4}} \quad (2)$$

where SR_B4 and SR_B5 denote shortwave bands 4 and 5, respectively.

NDVI values range from -1 to $+1$, where values closer to $+1$ indicate dense vegetation, while values closer to -1 represent urban areas or water bodies. After obtaining the NDVI, its minimum and maximum values over the study region are determined via regional reduction. These extreme values are then used to normalize the NDVI to derive the vegetation fraction (P_v). The normalization process involves subtracting the minimum NDVI and dividing by the range (max–min), and the result is squared to emphasize vegetation presence

$$P_v = \left(\frac{\text{NDVI} - \text{NDVI}_{\text{soil}}}{\text{NDVI}_{\text{veg}} + \text{NDVI}_{\text{soil}}} \right)^2 \quad (3)$$

where $\text{NDVI}_{\text{soil}}$ and NDVI_{veg} represent bare soil and vegetation NDVI, respectively.

The vegetation fraction is then employed in a linear equation to estimate the surface emissivity (ε), using the following equation:

$$\varepsilon = 0.004 \times P_v + 0.986 \quad (4)$$

where the values 0.004 and 0.986 indicate the slope coefficient and base emissivity value, respectively.

This parameter is critical because it adjusts the thermal measurements for ε variations, thereby improving the accuracy of LST calculations

$$\varepsilon_\lambda = \varepsilon_{\text{veg},\lambda} P_v + \varepsilon_{\text{soil},\lambda} (1 - P_v) + C_v \quad (5)$$

where ε_λ is the land surface emissivity at a specific wavelength; $\varepsilon_{\text{veg},\lambda}$ is the emissivity of a fully vegetated surface, typically ranging between 0.97 and 0.99; $\varepsilon_{\text{soil},\lambda}$ is the emissivity of bare soil, generally falling between 0.92 and 0.96; P_v is the fractional vegetation cover derived from NDVI; and C_v is a correction factor, usually around 0.005, that accounts for the effects of surface roughness and terrain irregularities on emissivity.

2) *Land Surface Temperature (LST) Calculation*: LST is a critical parameter for analyzing urban thermal variations and is derived from satellite TIR bands. The LST estimation follows these steps:

Top of atmosphere radiance is calculated as follows:

$$L_\lambda = \frac{(L_{\text{max}} - L_{\text{min}})}{Q_{\text{calmax}} - Q_{\text{calmin}}} \times (Q_{\text{cal}} - Q_{\text{calmin}}) + L_{\text{min}} \quad (6)$$

where spectral radiance (L_λ) represents the amount of energy emitted or reflected by a surface as recorded by the satellite sensor, measured in $\text{W/m}^2\text{sr}\mu\text{m}$. The spectral radiance is derived from the digital number (DN) value, denoted as Q_{cal} , which is the raw sensor reading. To convert DN values into physical radiance units, the calibration parameters, including the maximum and minimum spectral radiance values ($L_{\text{max}} - L_{\text{min}}$) and the corresponding maximum (Q_{calmax}) and minimum (Q_{calmin}) DN values, are required.

Brightness temperature is calculated as follows:

$$\text{BT} = \frac{K_2}{\ln \left(\frac{K_1}{L_\lambda} + 1 \right)} \quad (7)$$

where K_1 and K_2 are predefined thermal constants for the satellite sensor.

Finally, LST is computed as follows:

$$\text{LST} = \frac{\text{BT}}{1 + \lambda \cdot \text{BT} / \rho} \ln \epsilon \quad (8)$$

where λ denotes the wavelength of the emitted radiance. The term ρ represents Planck's function. The land EM, denoted by ϵ , accounts for the efficiency with which the surface emits thermal radiation relative to a perfect black body. It is calculated using an NDVI-based classification.

3) *Urban Heat Island (UHI) Effect*: The UHI effect, which measures the temperature difference between urban and rural areas, is a key indicator of urban-induced thermal variations. It is calculated as follows:

$$\text{UHI} = \frac{\text{LST} - \text{LST}_{\text{mean}}}{\text{LST}_{\text{std}}} \quad (9)$$

where LST is the LST value at a specific location, LST_{mean} represents the mean LST across the study area, and LST_{std} denotes the standard deviation of the LST. A higher UHI value indicates intensified urban heat and vice versa.

4) *Urban Thermal Field Variance Index (UTFVI)*: UTFVI integrates LST with vegetation indices (NDVI) to assess both human thermal discomfort and environmental degradation. It is calculated as follows:

$$\text{UTFVI} = \frac{\text{LST} - \text{LST}_{\text{mean}}}{\text{LST}} \quad (10)$$

where LST is the land surface temperature at a given location, and LST_{mean} is the mean land surface temperature across the entire study area. Higher UTFVI values indicate greater ecological stress and urban thermal discomfort. Following established literature [27], [67], the UTFVI in this article was classified into five ecological stress levels to enhance interpretability and facilitate spatial and temporal comparisons. UTFVI values less than 0 were considered to indicate “No Stress,” values between 0 and 0.05 indicated “Slight Stress,” values between 0.05 and 0.1 represented “Moderate Stress,” values from 0.1 to 0.15 were classified as “High Stress,” and values equal to or greater than 0.15 indicated “Extreme Stress.” These thresholds provided a basis for identifying thermal discomfort severity across the study area and supported the translation of remote sensing outputs into actionable categories relevant for urban ecological planning.

5) *Sen's Slope Estimator (SSE) Test*: SSE is a nonparametric trend analysis method used to measure the rate of change in urban thermal discomfort over time [68]. It is calculated as follows:

$$Q = \text{Median} \left(\frac{X_j - X_i}{j - i} \right) \forall i < j \quad (11)$$

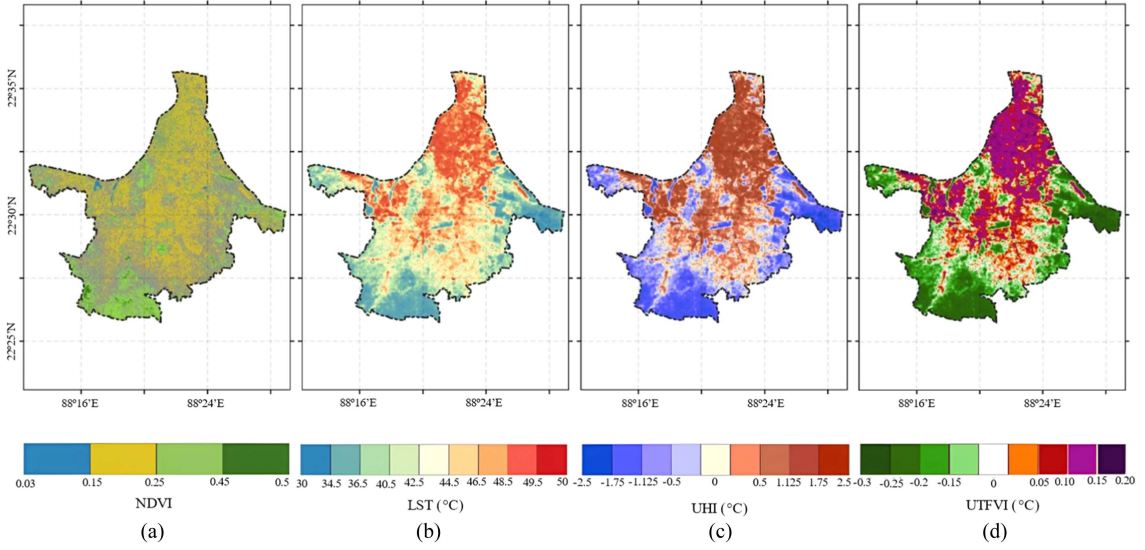


Fig. 2. Spatial distribution of long-term summer urban mean thermal discomfort indicators across Kolkata for the period 1995–2024: (a) NDVI, (b) LST, (c) UHI, and (d) UTFVI.

where X_i and X_j are the data values at times i and j , respectively, with i occurring before j . A positive Q value indicates an increasing trend (warming), while a negative Q value signifies a decreasing trend.

6) *Modified Mann–Kendall (mMK) Test*: The mMK test is a statistical method used to detect significant trends in thermal parameters while considering autocorrelation effects [69]. The test statistic “ S ” is calculated as follows:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(X_j - X_i) \quad (12)$$

where $\text{sgn}(X_j - X_i)$ is

$$\text{sgn}(X_j - X_i) = \begin{cases} +1, & X_j - X_i > 0 \\ 0, & X_j - X_i = 0 \\ -1, & X_j - X_i < 0. \end{cases} \quad (13)$$

The variance of S is given by

$$\text{Var}(S) = \frac{n(n-1)(2n+5) - \sum t_k(t_k-1)(2t_k+5)}{18} \quad (14)$$

where t_k represents tied ranks. The Z-score is computed as

$$Z = \begin{cases} \frac{S-1}{\sqrt{\text{Var}(S)}}, & \text{if } S > 0 \\ 0, & \text{if } S = 0 \\ \frac{S+1}{\sqrt{\text{Var}(S)}}, & \text{if } S < 0. \end{cases} \quad (15)$$

A positive Z-value indicates an increasing trend, while a negative Z-value signifies a declining trend. Statistical significance is evaluated at a 95% confidence level.

IV. RESULTS

A. Climatology of Urban Thermal Discomfort Indicators

The spatial distribution of summer mean NDVI, LST, UHI intensity, and UTFVI across Kolkata from 1995 to 2024 is presented in Fig. 2. As illustrated in Fig. 2(a), NDVI values across the Kolkata region exhibit spatial heterogeneity, ranging from 0.03 to ≥ 0.5 . The southern and southwestern peripheral zones display higher NDVI values (0.45–0.5), signifying areas of dense vegetation, urban parks, wetlands, and relatively undisturbed natural land. The central, northern, and eastern portions of the city record low NDVI values (0.03–0.25), indicative of sparse vegetative cover amid intense urban development. The patchiness of NDVI in the urban core reflects fragmented green spaces such as roadside vegetation, rooftop gardens, and institutional greenery.

Fig. 2(b) depicts the mean LST distribution from 1995 to 2024, revealing extreme heat accumulation in the central and northern sectors, where LST values consistently exceed 46 °C and approach 50 °C. These thermal hotspots correspond to zones of dense built-up areas, low vegetative presence, and high impervious surface fractions. In contrast, the southern, southwestern, and eastern fringes of the city exhibit cooler temperatures, with LST values ranging between 30 °C and 38 °C. These cooler patches coincide with zones of high NDVI, underscoring the cooling effects of vegetation through enhanced evapotranspiration and surface shading.

As shown in Fig. 2(c), the spatial climatology of UHI intensity ranges from -2.5 to $+2.5$ °C. Positive UHI values, particularly in the central, northeastern, and southeastern sectors (1 °C to 2.5 °C), highlight regions with severe urban thermal excess. These zones overlap with high LST and low NDVI regions, reflecting the compounded impact of vegetation loss and dense urbanization on thermal accumulation. Negative UHI values (-2.5 °C to -1 °C) in the southern, eastern, and western fringes

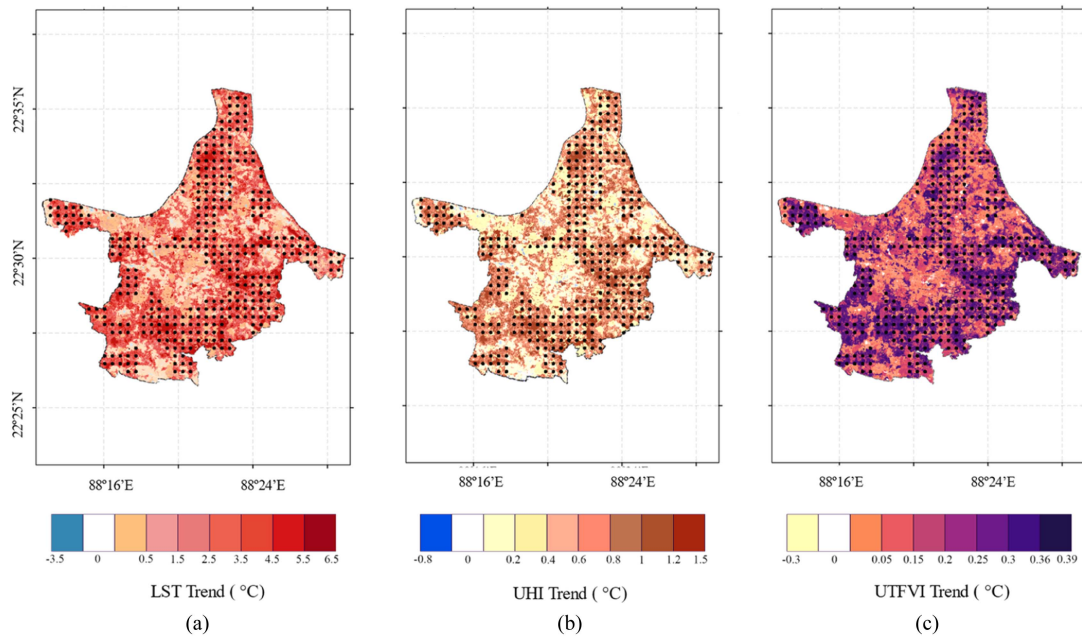


Fig. 3. Spatial trends of thermal discomfort indicators across Kolkata over the period 1995–2024: (a) LST, (b) UHI, and (c) UTFVI. Areas marked with black dots indicate statistically significant trends at the 0.05 significance level.

represent urban cool islands benefiting from vegetative and hydrological buffers.

Fig. 2(d) displays the UTFVI distribution, where index values range from ≤ -0.25 °C to ≥ 0.20 °C. High UTFVI values (0.15 °C to 0.20 °C) are concentrated in the central, northeastern, and selected southeastern localities, aligning spatially with LST and UHI hotspots. Meanwhile, areas in the southern periphery, particularly those adjacent to green infrastructure and water bodies, exhibit low to negative UTFVI values (≤ -0.15 °C), denoting zones of reduced ecological thermal stress and enhanced human thermal comfort.

B. Spatiotemporal Trends in Urban Thermal Discomfort Indicators

Fig. 3 shows the spatial distribution of the trend of thermal indicators over the period 1995–2024 in Kolkata. As illustrated in Fig. 3(a), the spatial trend in LST across Kolkata over the study period (1995–2024) ranges from -3.5 °C/year to $+6.5$ °C/year, with the majority of the city showing a significant increasing trend, especially in central, northern, southern, and eastern Kolkata.

Fig. 3(b) displays the trend in UHI intensity across Kolkata over the period 1995–2024. The results indicate that the trend values range from -0.8 °C/year to $+1.5$ °C/year. The northern, eastern, and southern sectors of the city exhibit the most intense and significant increase, with trend values exceeding $+1.0$ °C/year, signifying a rapid amplification of the UHI effect.

Fig. 3(c) illustrates the spatial trend in UTFVI across Kolkata for the period 1995–2024. The UTFVI trend values range from -0.3 °C/year to 0.39 °C/year, with widespread significant positive trends across the Kolkata area, particularly in the northeast,

central, and southeast zones. These regions display trend values exceeding $+0.25$ °C/year, which directly aligns with the mean UTFVI distribution. These regions already have high mean UTFVI values (0.15 °C/year to 0.20 °C/year), reflecting elevated thermal discomfort. The trend pattern confirms that these discomfort zones are not static—they are worsening significantly over time. These UTFVI trend hotspots closely correlate with the regions that have both high LST and UHI trends, suggesting that ecological stress is intensifying at the intersection of cumulative urban heat and poor vegetation cover.

C. Temporal Trends in Urban Thermal Discomfort Indicators

Fig. 4 shows the temporal trend of thermal indicators over the period 1995–2024 in Kolkata. As indicated in Fig. 4(a), the temporal trend of LST shows a consistent upward trajectory during the study period. The annual mean LST fluctuates year to year but maintains an overall increasing pattern, as indicated by the positive linear regression line. The calculated rate of increase is approximately 0.14 °C/year, confirming a persistent rise in LST over the study period. The temporal trend of LST in Kolkata exhibited distinct peak years across a total of 12 instances, specifically in 1996, 1999, 2001, 2003, 2006, 2009, 2012, 2014, 2016, 2019, 2021, and 2023, indicating recurring intervals of intensified surface heating over the study period.

The UHI trend demonstrates a similar, clear upward pattern over the study period. Despite year-to-year variability, as indicated in Fig. 4(b), the linear regression line shows a steady positive slope at the rate of 0.041 °C/year, suggesting that urban areas are becoming progressively warmer relative to surrounding rural areas. The UHI effect demonstrated peak intensities in

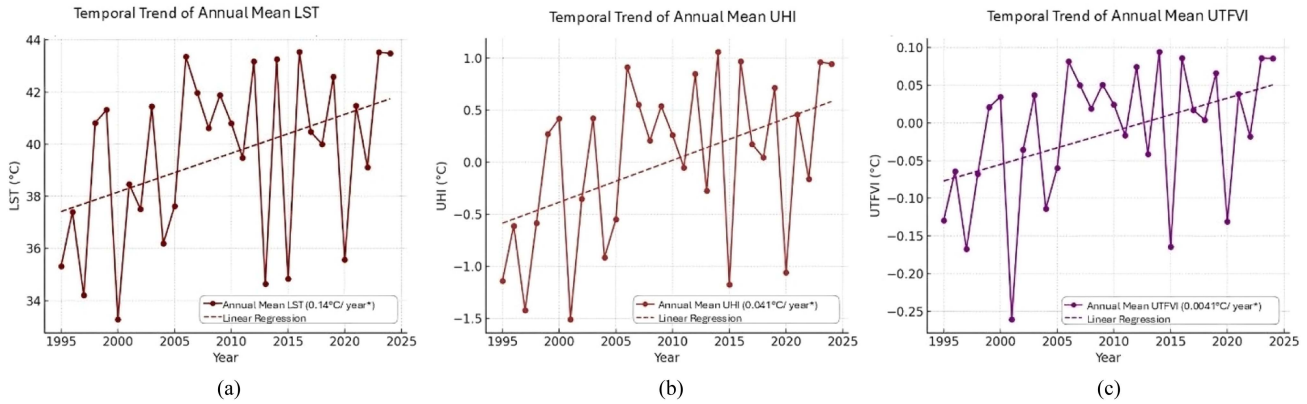


Fig. 4. Temporal trend of summer urban thermal discomfort indicators across Kolkata for the period 1995–2024: (a) LST, (b) UHI, and (c) UTFVI. The value in each panel indicates the monotonic trend, and the asterisk (*) indicates that the trend is statistically significant at the 0.05 significance level.

TABLE II
SPATIAL EXTENT OF THE UTFVI CATEGORIES FOR SELECTED YEARS

UTFVI category	Threshold range	Area in 1995 (km ²)	Area in 2005 (km ²)	Area in 2015 (km ²)	Area in 2024 (km ²)
No stress	UTFVI < 0	187.03	143.99	171.34	24.65
Slight stress	0–0.05	13.99	37.86	24.79	24.61
Moderate stress	0.05–0.10	2.74	17.76	6.66	64.35
High stress	0.10–0.15	0.17	4.23	0.57	64.90
Extreme stress	> 0.15	0.00	0.08	0.08	25.41
Total area	—	203.93	203.93	203.93	203.93

11 years, including 1996, 2000, 2003, 2006, 2009, 2012, 2014, 2016, 2019, 2021, and 2023, reflecting persistent and recurrent thermal amplification within the urban fabric of Kolkata.

Table II summarizes the spatial distribution of UTFVI-based ecological stress categories for the years 1995, 2005, 2015, and 2024. The results reveal a marked escalation in thermal stress across the study area. In 1995, approximately 91.7% of the total area (187.03 km²) was classified under the “No Stress” category (UTFVI < 0), indicating a thermally stable urban environment. However, by 2024, this category had alarmingly diminished to just 24.65 km², signifying a drastic reduction in thermally healthy zones. Concurrently, areas under “Moderate Stress” (UTFVI 0.05–0.10) and “High Stress” (UTFVI 0.10–0.15) witnessed a dramatic surge, from 2.74 and 0.17 km² in 1995 to 64.35 and 64.90 km², respectively, in 2024. The “Extreme Stress” category (>0.15), which was virtually nonexistent until 2005, expanded to 25.41 km² by 2024. These patterns clearly indicate an increasing prevalence of urban heat stress, likely linked to land cover change, loss of green spaces, and intensified urban development over the past three decades. This significant thermal shift underscores the urgent need for

sustainable urban planning and greening initiatives to mitigate UHI effects [70], [71], [72].

The UTFVI also exhibits an increasing trend from 1995 to 2024, as indicated in Fig. 4(c). The annual values show fluctuations but generally align with a positive regression line. The increasing rate of UTFVI is 0.0041 °C/year, signifying a gradual intensification of ecological thermal stress and thermal stress across the urban landscape of Kolkata. The UTFVI revealed 11 notable peak years, occurring in 1996, 2000, 2003, 2006, 2009, 2012, 2014, 2016, 2019, 2021, and 2023, suggesting consistent periods of heightened ecological thermal stress closely aligned with urban expansion trends.

Interannual variations in LST from 1995 to 2024 are influenced by both climatic variability and urban development patterns. The year 2000 in particular exhibits a relatively lower annual mean LST compared to surrounding years. However, this is accompanied by higher UHI and UTFVI values. This apparent discrepancy may result from increased temperature contrasts between urban and surrounding rural zones [73], [74], suggesting intensified localized heat accumulation even under relatively lower background temperatures. Additionally, a loss of vegetation cover in core urban zones between 1999 and 2001, corroborated by NDVI reductions, could have elevated UTFVI despite modest LST values [75], [76], [77]. This indicates that UHI and UTFVI are not solely dependent on absolute LST levels but are also shaped by spatial temperature gradients, land surface characteristics, and urban morphology [78], [79]. Furthermore, meteorological anomalies, such as increased cloud cover or premonsoon rainfall in certain years, may reduce overall LST but not eliminate UHI effects, which are computed as relative differences [80].

The monthly LST climatology and trend of Kolkata, as shown in Fig. 5(a), display the highest temperatures during April (44.64 °C) and May (44.15 °C), gradually decreasing through June (39.38 °C), July (37.90 °C), and reaching the lowest values in August (33.73 °C) before slightly increasing in September (37.66 °C). The trend analysis reveals positive trends in April (0.364 °C/year), May (0.351 °C/year), June (0.250 °C/year), and September (0.189 °C/year), while July (−0.022 °C/year)

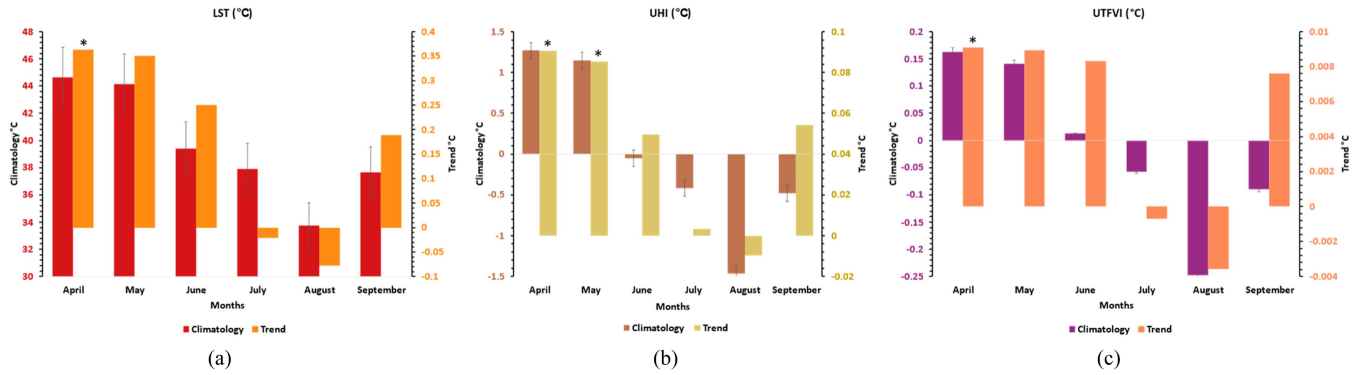


Fig. 5. Monthly trend of summer urban thermal discomfort indicators across Kolkata for the period 1995–2024: (a) LST, (b) UHI, and (c) UTFVI. The error bar indicates the standard deviation, while the asterisk (*) indicates that the trend is statistically significant at the 0.05 significance level.

and August (-0.078 °C/year) show slight negative trends. The largest positive trend is observed in April, indicating increasing surface temperatures, especially during the early summer season.

The monthly UHI climatology and trend of Kolkata, illustrated in Fig. 5(b), indicate that the UHI effect is strongest in April (1.27 °C) and May (1.14 °C), before turning negative in June (-0.05 °C), reaching its lowest in August (-1.46 °C), and moderately recovering in September (-0.48 °C). The trends show positive values for April (0.091 °C/year), May (0.085 °C/year), June (0.050 °C/year), July (0.003 °C/year), and September (0.054 °C/year), while August shows a minor negative trend (-0.010 °C/year). The most significant positive trend occurs in April, suggesting an intensifying UHI effect during the onset of the summer season.

The monthly UTFVI climatology and trend of Kolkata, depicted in Fig. 5(c), record positive climatology values in April (0.16), May (0.14), and June (0.01), while turning negative during July (-0.06), August (-0.25), and September (-0.09). Positive trends are evident in April ($0.0091/\text{year}$), May ($0.0089/\text{year}$), June ($0.0083/\text{year}$), and September ($0.0076/\text{year}$), suggesting increased ecological thermal stress. Meanwhile, slight negative trends are observed in July ($-0.0007/\text{year}$) and August ($-0.0036/\text{year}$). The highest positive trend again occurs in April, marking it as the peak period for worsening ecological thermal conditions.

V. DISCUSSION

This article aims to assess spatiotemporal changes in thermal discomfort in the urban environment of Kolkata, India, for the period 1995–2024. Urban expansion in rapidly growing megacities exerts a profound inverse impact on vegetative cover, a phenomenon consistently observed across Indian cities [81], [82], [83]. In Kolkata, the fragmentation of green spaces, including roadside vegetation, rooftop gardens, and isolated institutional plots, remains insufficient to regulate the urban microclimate or sustain ecosystem services, as similarly reported in assessments of Delhi and Bangalore cities [84]. This pattern highlights the urgent need for targeted greening interventions, such as urban forestry, vertical gardens, and green roof strategies, to restore ecological balance and mitigate surface heat accumulation in

densely built regions [85]. The long-term mean LST distribution from 1995 to 2024 reveals a cityscape increasingly shaped by urban morphology, vegetation loss, and land use change, aligning with spatial patterns documented in other Indian megacities [86], [87], [88].

The results show that the thermal hotspots dominate the central and northern zones of Kolkata, where dense construction, extensive impervious surfaces, and minimal vegetative cover exacerbate surface heating. In contrast, southern and peripheral zones, particularly those near wetlands and urban parks, exhibit comparatively lower LST values, reaffirming the moderating influence of vegetation and water bodies through evaporative cooling, a pattern also observed in the Indian cities of Tiruchirappalli and Hyderabad [84]. The pronounced spatial heterogeneity in LST illustrates how rapid and uneven urban expansion creates sharp microclimatic gradients across short distances. Temporal trend analysis of LST indicates an increasing trend, with the most intense warming concentrated in already heat-stressed areas, particularly the central and northeastern belts. This intensification has also been observed in the Indian cities of Delhi, Ahmedabad, and Bengaluru [90], [91]. These findings highlight the urgent necessity of incorporating surface temperature assessments into land use regulation, promoting interventions, such as reflective materials, ventilation corridors, and expanded green infrastructure, to counteract urban heat amplification [92].

The spatial distribution of UHI closely mirrors patterns observed in LST, with the central, northeastern, and southeastern sectors emerging as dominant heat island cores. These areas coincide with regions of high LST and low NDVI, reinforcing the compounded influence of dense built-up morphology, surface sealing, and vegetation loss; a similar pattern was also observed in other parts of the state of West Bengal, India [93], [94]. UHI trend analysis reveals an average increase of 0.041 °C per year, with the steepest intensification observed in the central and eastern areas. The monthly UHI climatology and trend of Kolkata reveal the strongest UHI effect in April and May, with generally positive trends across most months, highlighting an intensifying effect of UHI during the early summer season. This pattern confirms that UHI hotspots are worsening, driven by ongoing urbanization and surface warming, consistent with findings from the state capital cities of Ahmedabad, Delhi, and Imphal [95],

[96], [97]. Strategic planning measures, including canopy expansion, albedo management, and cooling corridor development, are critical to addressing these vulnerabilities [98], [99].

The UTFVI spatial distribution complements these findings by integrating ecological and human-centered dimensions of thermal stress. High UTFVI values are concentrated in zones affected by elevated LST and intensified UHI effects, confirming patterns previously reported in the Prayagraj, Asansol, and Chotanagpur regions of India [100], [101]. Conversely, zones near major green infrastructures and water bodies exhibit lower UTFVI values. Temporal analysis reveals that UTFVI increased at $0.0041\text{ }^{\circ}\text{C}$ per year, closely tracking LST and UHI trends, and emphasizing the escalating vulnerability of the urban population. The monthly UTFVI climatology and trend of Kolkata reveal positive values and rising trends in most months, with April marking the peak of increasing ecological thermal stress. These results align with the findings of the previous articles, which highlight urban densification, surface sealing, and vegetation loss as primary drivers of rising urban heat in Indian cities [48], [102], [103]. The spatial convergence of high thermal values with low NDVI zones reinforces the inverse relationship between vegetation cover and thermal stress, a pattern observed in several other tropical megacities [66], [104].

The methodological framework adopted here, spanning remote sensing, ecological indices, and nonparametric statistics, offers a replicable model for other rapidly urbanizing cities confronting climate-related thermal stress [105], [106]. In line with existing studies that have documented heat stress amplification in rapidly urbanizing Indian cities [107], [108], this article advances the methodological framework by integrating cloud-based remote sensing, long-term trend diagnostics, and spatial significance mapping across multiple indices [65], [109]. The inclusion of UTFVI, often overlooked in urban heat studies, provides a critical ecological perspective, linking thermal changes to the broader urban environment and human comfort [27], [55].

These findings call for multidimensional urban interventions. Greening initiatives should be prioritized in thermal hotspots identified as both statistically significant and ecologically vulnerable. Urban planning must integrate cool roof technologies, green corridors, and climate-sensitive zoning, particularly in areas undergoing land-use transition [3], [110]. As the thermal landscape of Kolkata continues to evolve, such data-driven strategies will be essential for building urban resilience and ensuring sustainable human–environment interaction in the face of growing climatic pressures.

Though the study adopted a holistic approach and reported significant results, this article is limited by its geographic focus on a single Indian city, Kolkata, which may restrict the broader applicability of the findings to other urban environments. Moreover, it relies solely on remote sensing data without incorporating ground-based data, which could affect the efficacy of the derived thermal indices. Additionally, the use of multiple satellite sensors (Landsat-5, Landsat-7, and Landsat-8) introduces potential inconsistencies in the data due to differences in sensor characteristics. Future articles should aim to incorporate ground-based in-situ measurements, expand the analysis to

include multiple cities, integrate socioeconomic variables to better understand human vulnerability, and develop future projections under different urban growth and climate change scenarios.

While the findings of this article offer valuable insights into the spatial and temporal dynamics of urban thermal discomfort in Kolkata, one important limitation is the absence of ground-based validation for LST, UHI, or UTFVI estimates. Although satellite-derived thermal indices have been extensively validated in other urban studies [60], [111], the lack of long-term, station-level temperature records in Kolkata restricts the opportunity for local calibration. This challenge is common in many rapidly urbanizing regions of the Global South. Future article should aim to integrate in situ observations from meteorological stations, mobile transect surveys, or urban sensor networks to improve the accuracy of thermal assessments. Additionally, the use of reanalysis datasets such as ERA5-Land or MODIS-derived ground-validated LST could serve as proxy references in the absence of local data. Incorporating such validation frameworks would enhance the credibility of satellite-based urban thermal metrics and allow for better policy translation.

The use of multisensor Landsat data (TM, ETM+, and OLI) from Landsat 5, 7, and 8 introduces potential inconsistencies in radiometric response and spectral calibration. To reduce these effects, we applied uniform preprocessing workflows across all sensors using GEE, including radiance-to-temperature conversions, NDVI computation, and emissivity correction, following well-established Landsat LST and vegetation index models. Additionally, by focusing on long-term median trends and using nonparametric statistical methods (SSE and mMK), we minimized the influence of year-to-year sensor discrepancies. Nevertheless, we acknowledge that an explicit intersensor normalization or quantitative uncertainty analysis could further enhance trend reliability. Future article could adopt a cross-calibration technique, such as radiometric harmonization models, to refine uncertainty bounds in UTFVI trend assessments.

Another limitation of this article is the absence of spatial analysis linking urban thermal discomfort with socially vulnerable populations, such as slum dwellers, elderly residents, and economically marginalized groups. While the UTFVI maps reveal spatial patterns of thermal stress, they do not directly account for the differential exposure, sensitivity, or adaptive capacity of these communities. This is particularly critical in a megacity like Kolkata, where high-density informal settlements often coincide with low vegetation cover and poor housing materials, exacerbating heat exposure risks. The lack of high-resolution demographic data constrained the ability to conduct spatial overlay analysis in this article. However, future article should integrate gridded population datasets, ward-level census indicators, or vulnerability indices to identify hotspots where urban heat intersects with social inequity. Such integration would not only enhance the scientific rigor of urban thermal assessments but also support more equity-centered urban climate adaptation policies.

VI. CONCLUSION

This article presents a comprehensive spatiotemporal assessment of urban thermal discomfort in Kolkata from 1995 to 2024,

leveraging a cloud-based GEE platform with high-resolution remote sensing and robust statistical techniques. By integrating urban thermal indicators, such as LST, UHI, and the UTFVI, with vegetation dynamics (NDVI), the research provides critical insights into the evolving thermal landscape of Kolkata, one of India's fastest growing metropolitan regions. The results confirm a statistically significant warming trend, with LST increasing at a rate of $+0.149^{\circ}\text{C}/\text{year}$, UHI at a rate of $+0.041^{\circ}\text{C}/\text{year}$, and UTFVI at a rate of $+0.0041^{\circ}\text{C}/\text{year}$. These changes are spatially concentrated in high-density urban cores, particularly the central and northeastern Kolkata zones, which also report low NDVI values, signifying severe vegetation loss and impervious surface expansion, possibly due to large-scale conversion of wetland and cultivated land into built-up area.

This article provides new evidence on the evolution of urban thermal discomfort in Kolkata over three decades, using UTFVI and satellite-derived LST-NDVI trends on the GEE platform. By integrating nonparametric trend analysis with UTFVI classification, the study offers a scalable and cloud-efficient framework for long-term thermal monitoring in other Indian megacities experiencing rapid urbanization. The observed rise in extreme thermal stress zones, particularly in densely built-up and vegetation-sparse areas, underscores the urgency of targeted heat mitigation. Future article should combine remote sensing with fine-scale demographic and infrastructure data to inform community-level resilience strategies, especially for vulnerable populations in informal settlements. These insights can guide adaptive urban planning, green infrastructure deployment, and climate-resilient policy interventions in India's high-risk urban corridors.

The study findings recommend that future article should incorporate higher resolution datasets and socioeconomic variables to better understand localized thermal impacts and vulnerable populations. Integrating ground observations and seasonal analysis can also enhance the efficacy of thermal discomfort assessments. From a policy perspective, prioritizing green infrastructure, ecological zoning, and climate-sensitive design is essential. Embedding such data-driven approaches into urban planning will be key to fostering climate resilience in Kolkata and similar rapidly urbanizing cities.

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Data Availability: The remote sensing data can be accessed at earthexplorer.usgs.gov and earthengine.google.com, while codes can be provided upon a formal request to the corresponding author.

Conflict of Interest Statement: The authors declare no conflicts of interest.

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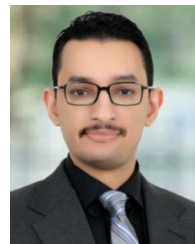
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